KEY ANSWERS

UNIT-8

CHEMICAL EQUILIBRIUM

EVALUATE YOURSELF

1.

	Fe^{3+}	SCN^-	$[Fe(SCN)^{2+}$
Initial Concentration(M)	1×10^{-3} (10×10^{-4})	8×10^{-4}	–
Reacted	2×10^{-4}	2×10^{-4}	–
Equilibrium Concentration	8×10^{-4}	6×10^{-4}	2×10^{-4}

$$K_{eq} = \frac{[Fe(SCN)]^{2+}}{[Fe^{3+}][SCN^-]}$$

$$= \frac{2\times10^{-4}\ M}{8\times10^{-4}\ M \times 6\times10^{-4}\ M}$$

$$= 0.0416\times10^{4}$$

$$K_{eq} = 4.16\times10^{2}\ M^{-1}$$

2. $2\ NO(g) + O_2\ (g) \rightleftharpoons 2NO_2(g)$

	NO_2	O_2	NO_2
Initial Partial Pressure	1	1	–
Reacted	0.96	0.48	–
Equilibrium Partial Pressure	0.04	0.52	0.96

$$K_P = \frac{P^2_{NO_2}}{P^2_{NO}\ .\ Po_2}$$

$$= \frac{0.96\times0.96}{0.04\times0.04\times0.52}$$

$$= 11.07\times10^{2}\ (atm)^{-1}$$

3. $CO(g) + H_2O(g) \rightleftharpoons CO_2\ (g) + H_2(g)$

Given $K_P = 2.7$

$[CO] = 0.13$, $[H_2O] = 0.56$

$[CO_2] = 0.78$; $[H_2] = 0.28$

$V = 2L$

$K_P = K_C\ (RT)$

$2.7 = K_C\ (RT)^o$

$K_C = 2.7$

$$Q_C = \frac{[CO_2][H_2]}{[CO][H_2O]}$$

$$= \frac{\left(\frac{0.78}{2}\right)\left(\frac{0.28}{2}\right)}{\left(\frac{0.13}{2}\right)\left(\frac{0.56}{2}\right)}$$

$Q = 3$

$Q > K_C$, Hence the reaction proceed in the reverse direction.

4. $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Given that $[PCl_5]_{initial} = 1$ mol; $V = 1\ dm^3$; $K_C = 2$

	PCl_5	PCl_3	Cl_2
Initial no. of moles	1	–	–
No. of moles	x	–	–
No. of moles at equilibrium	1–x	x	x
Equilibrium concentration	$\frac{1-x}{1}$	$\frac{x}{1}$	$\frac{x}{1}$

$$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$$2 = \frac{x \times x}{(1-x)}$$

$2 - 2x = x^2$

$x^2 + 2x - 2 = 0$

solution for a quadratic equation $ax^2 + bx + c = 0$ are,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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